

roles are positions, not words: every template token has its own role, the interior of the user turn shares one, the suffix is indexed from the end. 'dog' and 'cat' are z-scored against the same clean (layer, role) cell wherever they sit in the interior

(a) user message 'Can my dog eat cat food?': 36 tokens after Qwen's chat template, role above, token below

< im_start >	pre0	pre1	pre2	pre3	pre4	pre5	pre6	pre7	pre8	pre9	pre10	pre11	pre12	pre13	pre14	pre15	pre16	pre17	pre18	pre19	pre20	pre21	pre22	pre23	content_first	content_second	content	content	content	content	content_last	suf-5	suf-4	suf-3	suf-2	suf-1							
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35								
system	\n		You	are	Q	wen	,	created	by	Alibaba	Cloud	.	You	are	a	helpful	assistant	.	< im_end >		\n		< im_start >		user	\n		Can	my	dog	eat	cat	food	?	< im_end >		\n		< im_start >		assistant	\n	

(b) user message 'Can my cat eat dog food, and is it safe long term?': 43 tokens after Qwen's chat template, role above, token below

< im_start >	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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pre0	pre1	pre2	pre3	pre4	pre5	pre6	pre7	pre8	pre9	pre10	pre11	pre12	pre13	pre14	pre15	pre16	pre17	pre18	pre19	pre20	pre21	pre22	pre23	content_first	content_second	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content	content

template prefix: one role per position, indexed from the start

user-turn edges: content_first, content_second, content_last

interior content: one shared role

template suffix: one role per position, indexed from the end